GlobPOP accurately predicts global population trends, **proving reliable even in developing countries**.

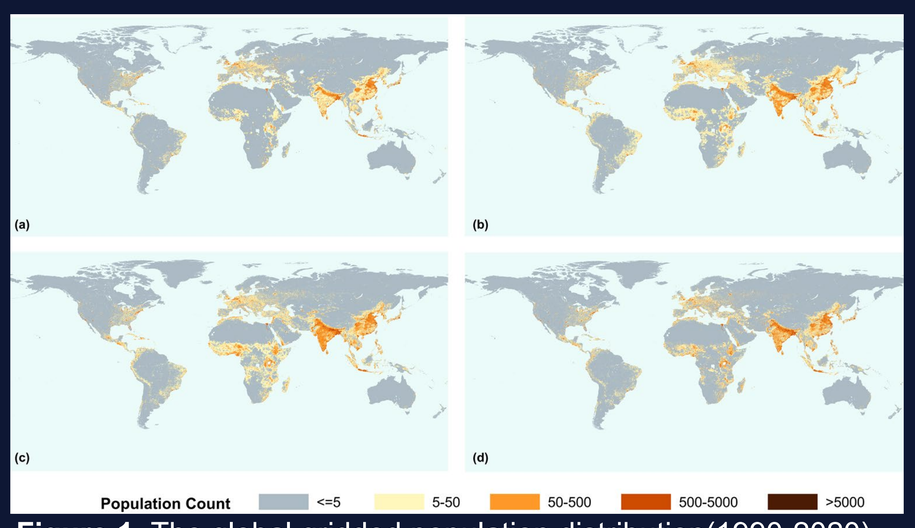

Figure 1. The global gridded population distribution(1990-2020)

GlobPOP: A 31-year (1990-2020) global gridded population dataset generated by cluster analysis and statistical learning

Luling Liu[1,2], Xin Cao[1,2], Shijie Li[1], Na Jie[1]

Background

Existing global gridded population datasets suffer from limitations such as a lack of continuous long-term data at high spatial resolution and uncertainties related to methodological and ancillary data.

Objective

To integrate these five sets of multi-source demographic data and leverage the strengths of each data through a statistical learning approach, and produce a set of new demographic products suitable for long time-series analysis at the grid scale.

Result 1

The graph illustrates the remarkable accuracy of GlobPOP in predicting country-level population estimates comparing with the World Population Prospects 2022.

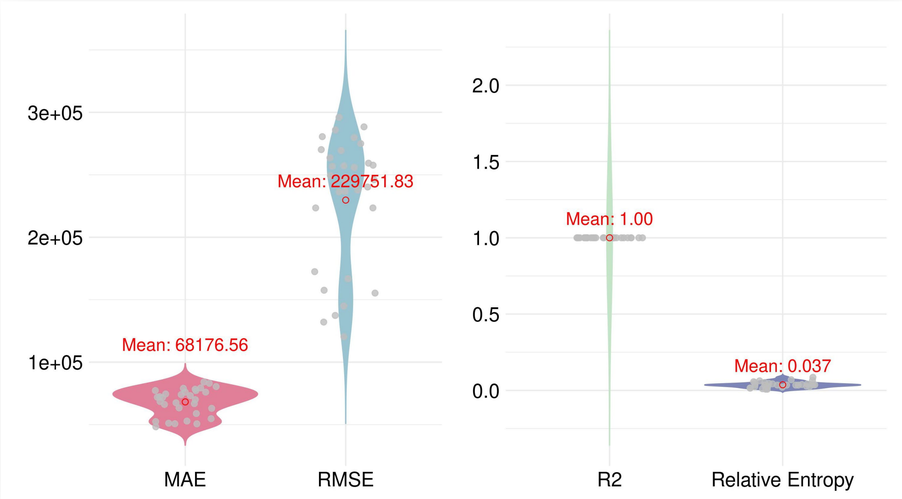

Figure 2. Spatial accuracy at the level-0 scale from 1990 to 2020

Result 2

The figure compares population time-series curves for eight countries using the GlobPOP dataset and five others, highlighting GlobPOP's superior temporal accuracy. Dynamic Time Warping(DTW) distances reinforce this, consistently showing GlobPOP outperforming other datasets across all eight countries.

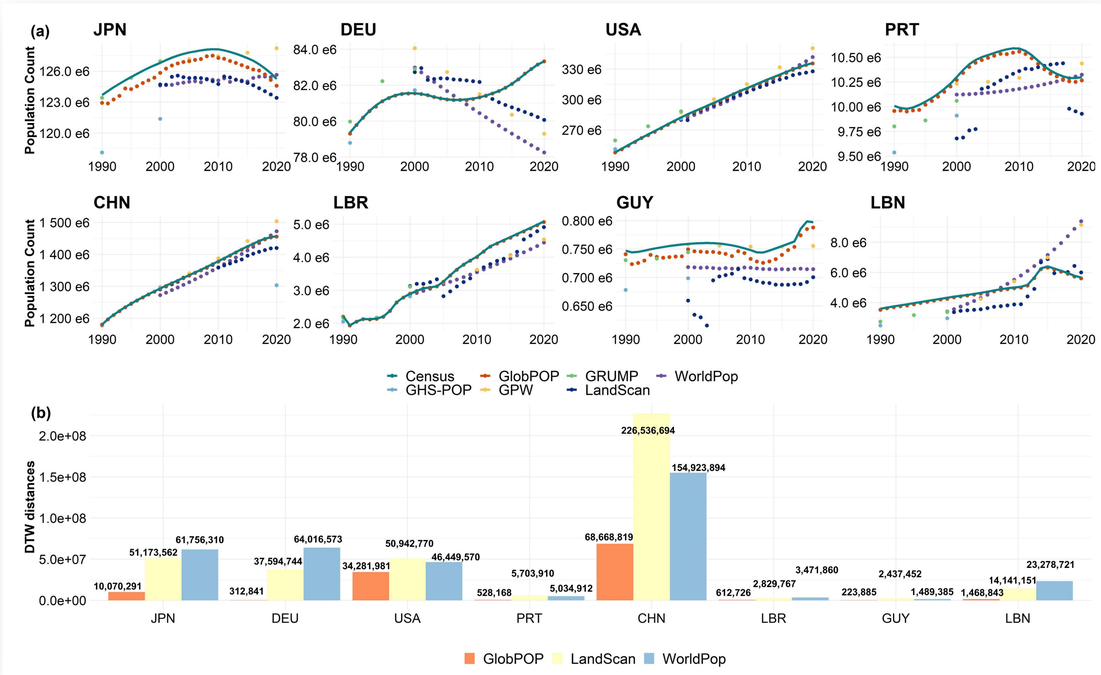

Figure 3. Comparison of the GlobPOP and other datasets over eight countries

Methods

This study introduces a novel data fusion framework to create a continuous global gridded population dataset, GlobPOP, spanning from 1990 to 2020.

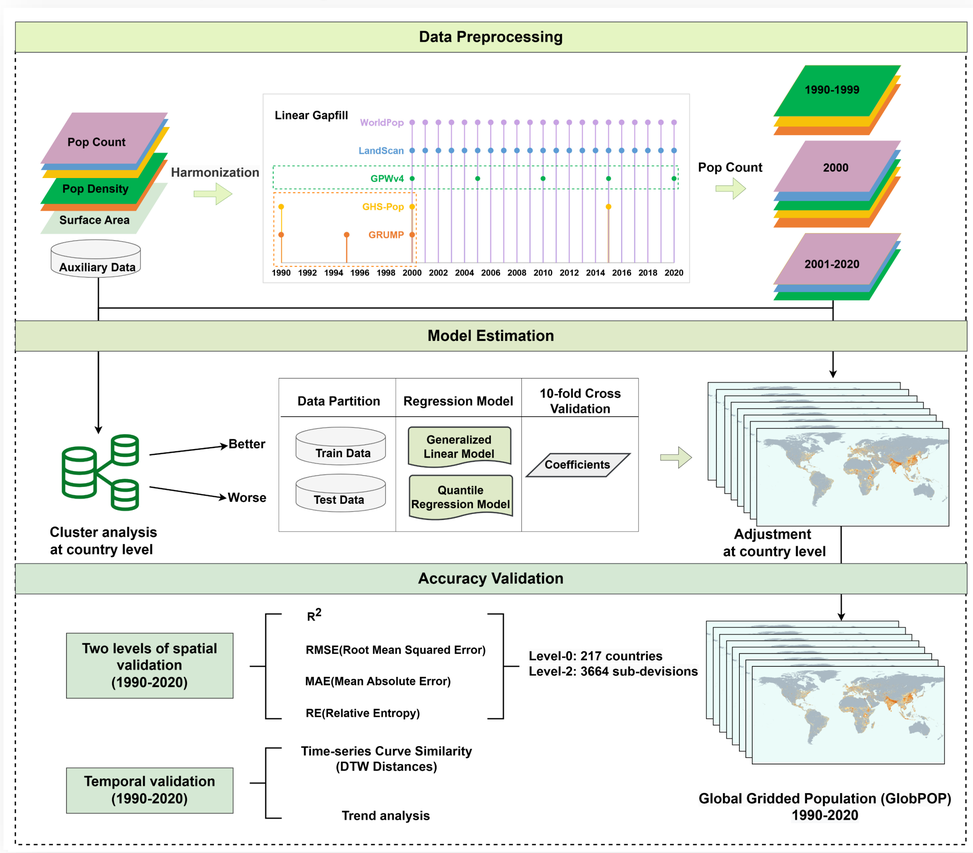

Figure 4. Workflow of the estimation and validation of the global gridded population (GlobPOP)

Usage notes

Researchers and policymakers can leverage the new dataset to conduct time-series analysis of the population and explore the spatial patterns of population development at global, national, and city levels.

Acknowledgments

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Affiliations

1. State Key Laboratory of Remote Sensing Science, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China
2. Beijing Engineering Research Center for Global Land Remote Sensing Products, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China

Contact:
lulingliu@mail.bnu.edu.cn